

# Dalton Kutzen

## Curriculum Vitae

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### Education

**Brigham Young University — B.S. Molecular Biology**

Jan 2025 – Apr 2028

[Provo, Utah](#)

- Coursework: molecular biology, genetics, statistics, calculus, bioinformatics, machine learning, and systems biology.
- Focused on computational approaches to genomics and reproducible biological research.

### Experience

**Bioinformatics Researcher**

Sep 2025 – Present

[Terrooatea Lab, Brigham Young University](#) · Provo, UT

- Developed computational methods and benchmark evaluations for high-dimensional single-cell RNA-seq (scRNA-seq) and pooled Perturb-seq screens (anchor-op).
- Investigated whether linear steady-state models with additive-input encodings generalize to operator-level held-out targets across Replogle and Jost datasets.
- Implemented target-grouped nested cross-validation and matched signal-to-noise ratio (SNR) controls in Python to evaluate prediction performance against baseline models.
- Co-authored bioRxiv research on perturbation screen modeling and computational reproducibility.

**Research Assistant**

Sep 2025 – Present

[Grose Phage Lab, Brigham Young University](#) · Provo, UT

- Prepared media, cultured bacteriophages with bacterial hosts, and conducted plaque and spot assays to evaluate phage activity and host interactions.
- Performed bioinformatic analyses to annotate and compare phage genomes, creating the **Phantom** Snakemake pipeline for reproducible annotation.
- Integrated wet lab microbiology techniques with computational genomics to investigate phage diversity, host range, and therapeutic applications.

**IT Service Desk Supervisor**

May 2026 – Present

[Brigham Young University — College of Life Sciences](#) · Provo, UT

- Lead frontline IT support for faculty, staff, and students across the College of Life Sciences, resolving high-complexity hardware, software, and systems incidents.
- Mentor student technicians in troubleshooting best practices and train new hires on Life Sciences IT infrastructure, tools, and clear user communication.
- Drove internal automation, documentation, and reporting initiatives that reduced repetitive workflows and standardized support delivery college-wide.

**IT Service Desk Technician**

Aug 2025 – May 2026

[Brigham Young University](#) · Provo, UT

- Oversaw deployment, security, and maintenance of 1,000+ campus workstations and lab computers, minimizing system downtime.
- Supported a community of 5,000+ users by troubleshooting hardware, software, and wireless networking issues for academic and administrative operations.

**Systems & Infrastructure Projects**

Ongoing

Personal Homelab · Provo, UT

- Administer Linux servers, virtual machines, and containerized services using Proxmox and self-hosted infrastructure.
- Designed practical monitoring, networking, backup, and automation workflows for resilient compute environments.

### Labeling and Packaging Specialist

Sep 2024 – Dec 2024

Uintah Basin Herbals · Vernal, UT

- Ensured strict compliance with Good Manufacturing Practice (GMP) standards for accurate labeling and packaging of herbal products.
- Collaborated with cross-functional production teams to maintain high throughput and meet scheduled distribution deadlines.

### Sales Representative

Apr 2024 – Jul 2024

OptimalShip · Rexburg, ID

- Maintained precise customer records and data integrity within CRM systems for reliable client tracking and follow-up.
- Managed and prioritized inbound and outbound client communication during peak operational hours.

## Publications & Preprints

- Fullmer, K. J., **Kutzen, D.**, & Terrooatea, T. W. (2026). Additive-input encoding fails operator-level target-held-out prediction on current Perturb-seq screens. *bioRxiv*. [Code]

## Presentations & Posters

- **Kutzen, D.** (2025). *Comparative Genomics and Automated Annotation of Novel Bacteriophages*. Poster Presentation, Life Sciences Undergraduate Research Symposium, Provo, UT.

## Selected Projects

### Phantom — Bacteriophage Analysis Workflow

2025 – Present

- Modular Snakemake pipeline for phage genome annotation and downstream comparative analysis.
- Uses isolated Conda environments and declarative workflow rules to make runs reproducible across systems.
- Produces organized intermediate files, results, logs, and publication-oriented visualizations.

### Anchor-Op — Perturb-seq Operator Benchmark

2026

- Computational evaluation suite assessing whether additive-input encodings generalize to held-out perturbation targets on pooled Perturb-seq screens.
- Implemented target-grouped nested cross-validation and matched-SNR controls across large-scale single-cell functional genomics datasets.

### Phage Visualization Tools

2025

- Developed R and Python utilities for post-annotation exploration of bacteriophage genomes.
- Designed figures and tables to help interpret gene content, genome architecture, and comparative results.

### Self-Hosted Research Infrastructure

Ongoing

- Built a Linux-based environment for code hosting, containers, services, and research project organization.
- Applied infrastructure-as-code and documentation practices to reduce configuration drift.

## Technical Skills

| Area                        | Tools & Proficiencies                                                                                     |
|-----------------------------|-----------------------------------------------------------------------------------------------------------|
| Computational Biology       | Comparative genomics, phage genome annotation, Perturb-seq analysis, Snakemake, Conda, Biopython          |
| Wet Lab & Molecular Biology | Phage culturing, plaque & spot assays, media preparation, agarose gel electrophoresis, host range testing |

| Area                                   | Tools & Proficiencies                                                                                                                         |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming &amp; Data</b>          | Python, R, Bash, SQL, Git, pandas, tidyverse, ggplot2, Quarto                                                                                 |
| <b>IT &amp; Systems Infrastructure</b> | IT Service Management (ITSM), Linux server administration, Proxmox, Docker, 1,000+ endpoint management, networking                            |
| <b>Operations &amp; Leadership</b>     | Good Manufacturing Practice (GMP), project management, technical mentoring, volunteer training (200+ individuals), CRM systems, documentation |

## Leadership & Service

### Initiative Lead

Jan 2026 – Present

[Leadership and the Scriptures](#), Brigham Young University · Provo, UT

- Plan and facilitate structured discussions connecting scriptural principles with practical leadership in academic, community, and professional settings.
- Guide participants in developing actionable leadership habits centered on empathy, honoring agency, and fostering personal and interpersonal accountability.

### Counselor and Youth Mentor

May 2025 – Aug 2025

[FSY Conferences](#) · United States

- Mentored cohorts of 15–25 youth participants, facilitating daily meetings, leadership workshops, and goal-setting activities.
- Motivated participants to establish actionable personal and academic goals, develop effective study habits, and strengthen character traits.
- Reported regularly to conference leadership regarding participant progress, group dynamics, and logistical improvements.

### Full-Time Volunteer Representative & Trainer

Mar 2022 – Mar 2024

[The Church of Jesus Christ of Latter-day Saints](#) · Kansas

- Supervised and trained 200+ volunteers throughout the state of Kansas, enhancing operational productivity and organizational goal execution.
- Provided daily instruction, counseling, and community service to individuals and families to support personal well-being and life skills.

### Civic Engagement Student Representative

Feb 2025 – Present

[Brigham Young University](#) · Provo, UT

- Participated in municipal policy and civic outreach initiatives with Provo City leadership.
- Advocated for student civic participation, local governance engagement, and community voting awareness.

## Honors & Awards

- **Dean's List / Academic Recognition**, Brigham Young University

2025